

End-to-End Data Engineering Pipeline for HR Employee Attrition Analysis

Data Engineering

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Data Modelling (Star Schema)

Data Transformation Flow

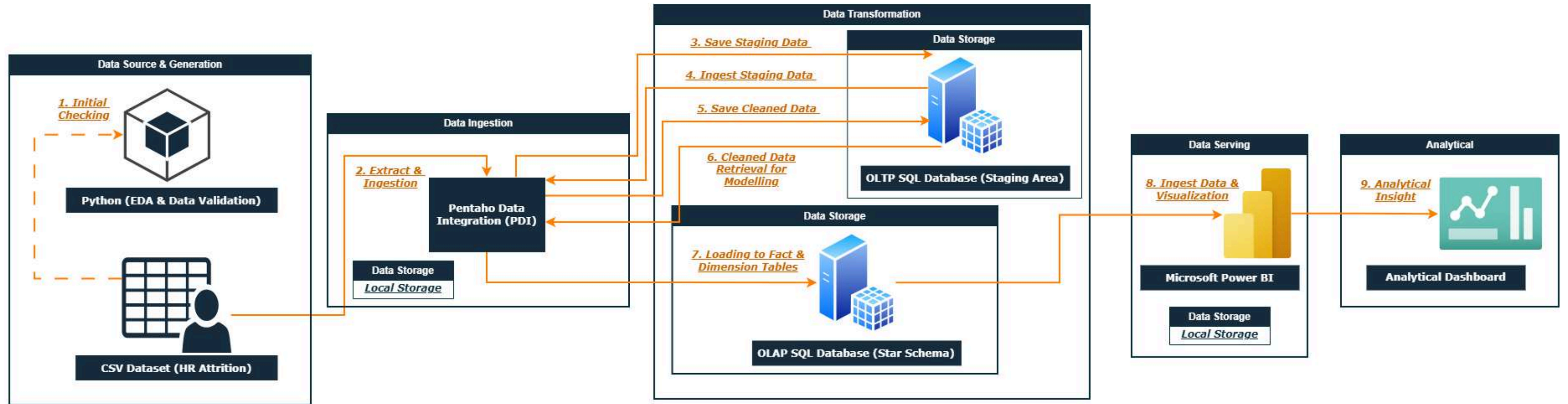
Orchestration with Pentaho Jobs

Data Consumption (Power BI)

Dataset Information

- **Personal Data:** consists of basic information such as Age, Gender, Marital Status, Education Field, Education, Monthly Income, Monthly Rate, Daily Rate, Hourly Rate, Percent Salary Hike, Stock Option Level, Over 18, Employee Number, and Employee Count.
- **Work History & Status:** contains information about the job such as Job Title, Department, Job Level, Business Travel, Distance From Home, Total Working Years, Number of Companies Worked, Years at the Company, Years in Current Role, Years Since Last Promotion, Years with Current Manager, Training Times Last Year, Overtime, and Standard Hours.
- **Job Satisfaction, Environment & Retention:** contains employee assessments of the Environment, Job Involvement, Relationship Satisfaction, Work-Life Balance, Job Satisfaction, Performance Rating, and Attrition.

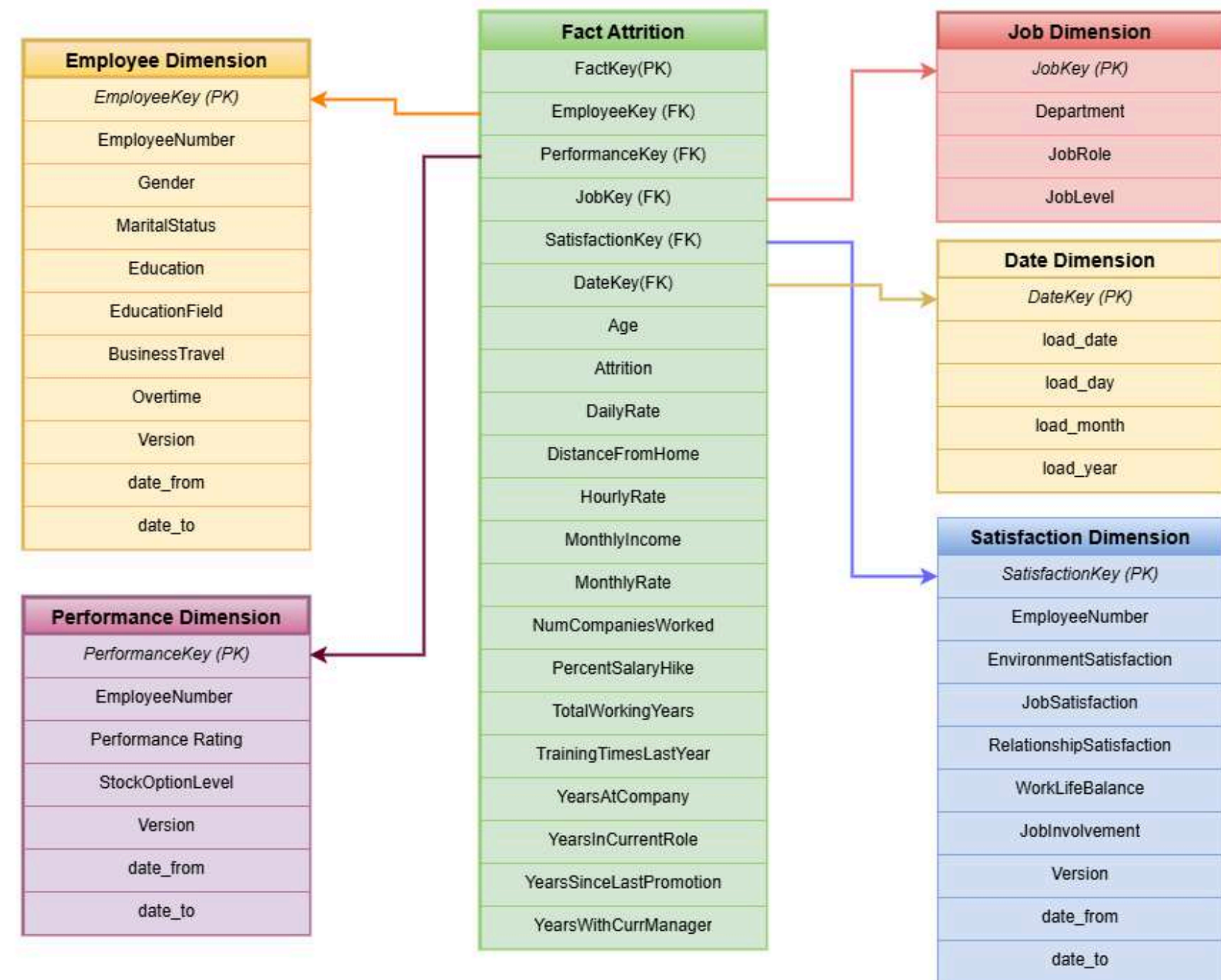
Project Flow / Framework



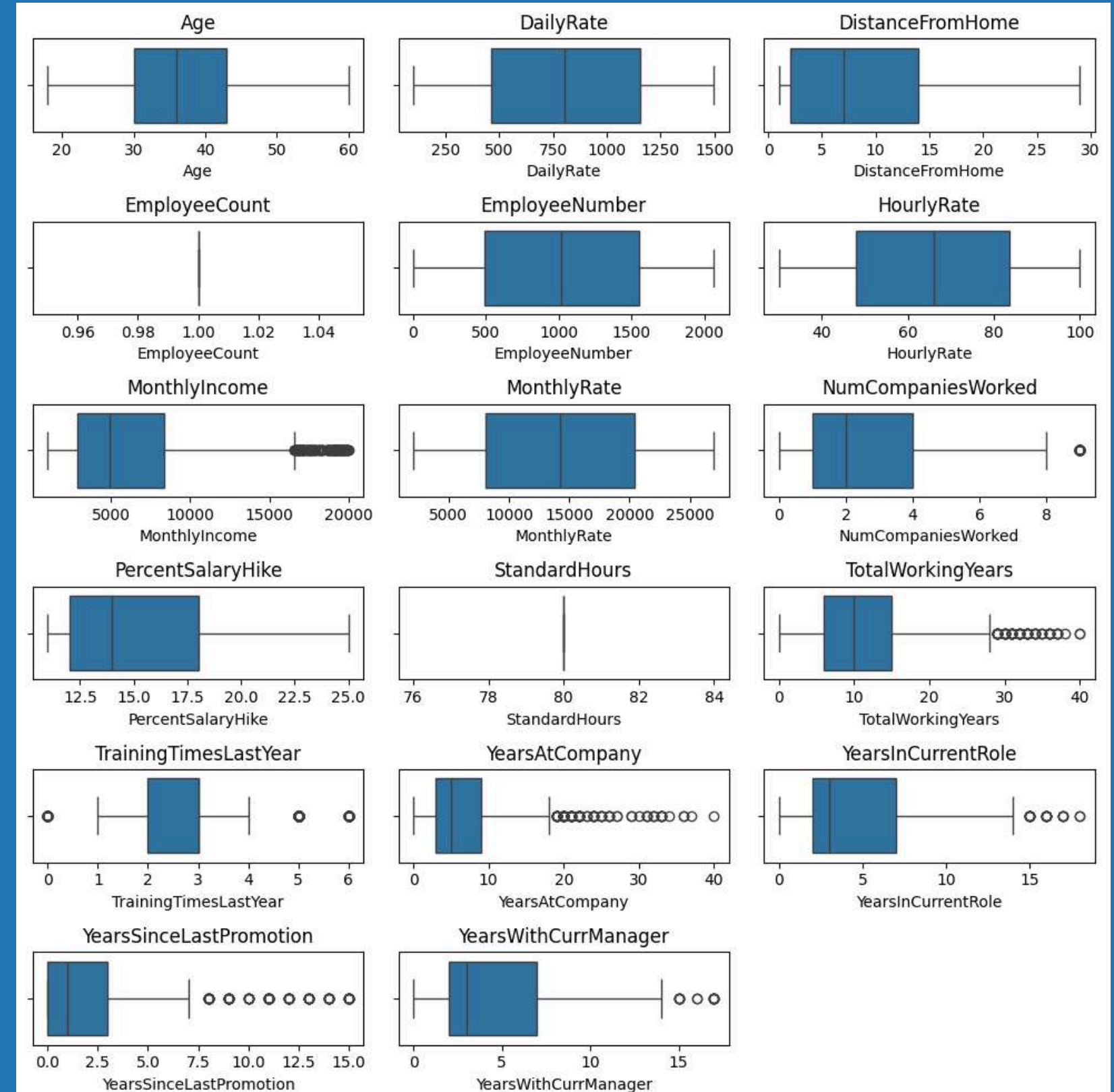
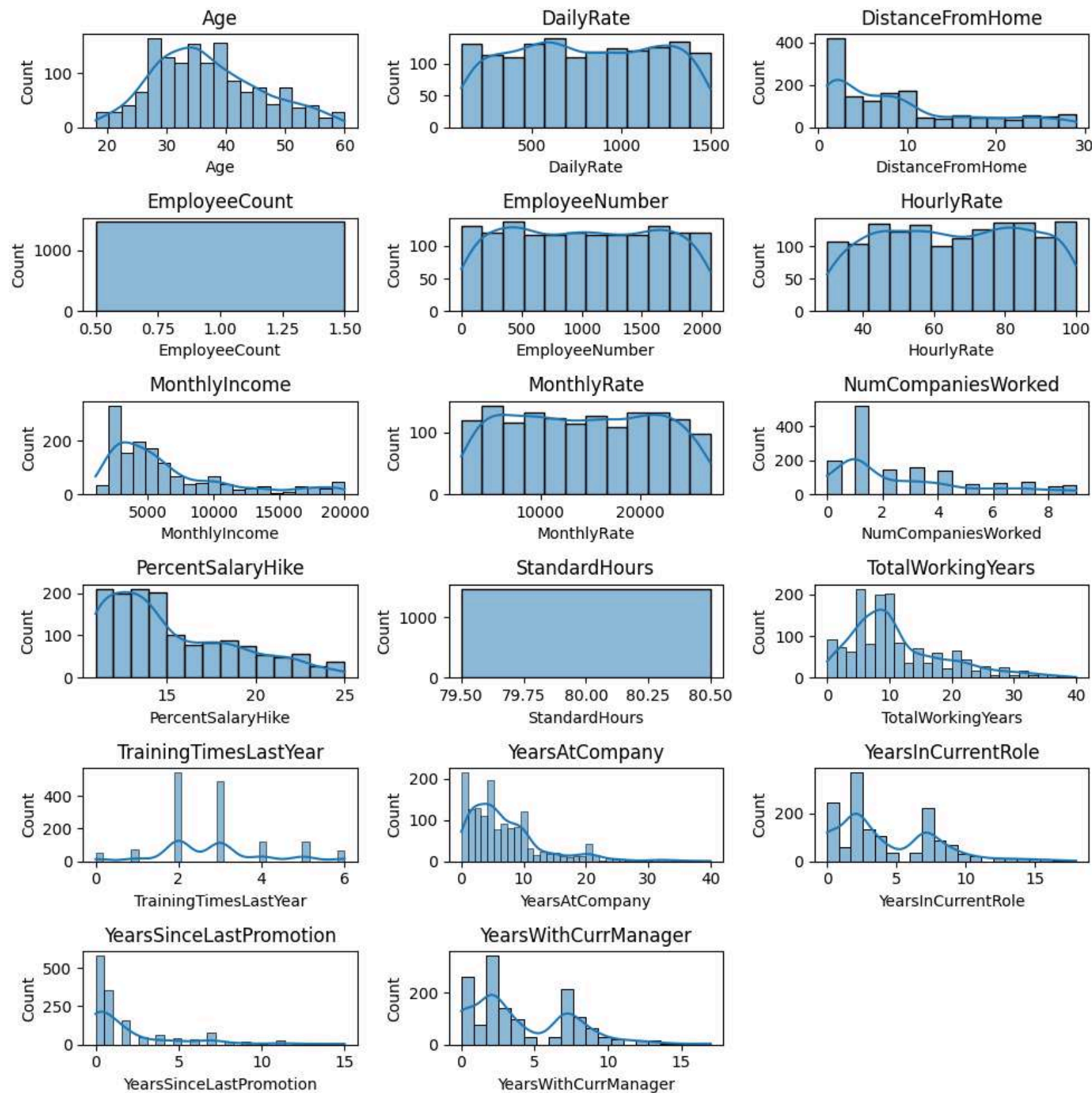
Data Modelling

- Fact Table: Menyimpan data measurement yang membutuhkan perhitungan utk dianalisis.
- Dimensions: Dim_Employee, Dim_Job, Dim_Satisfaction, Dim_Performance, & Dim_Date
- Benefit: Desain ini mempercepat proses filtering dan slicing data di Power BI.

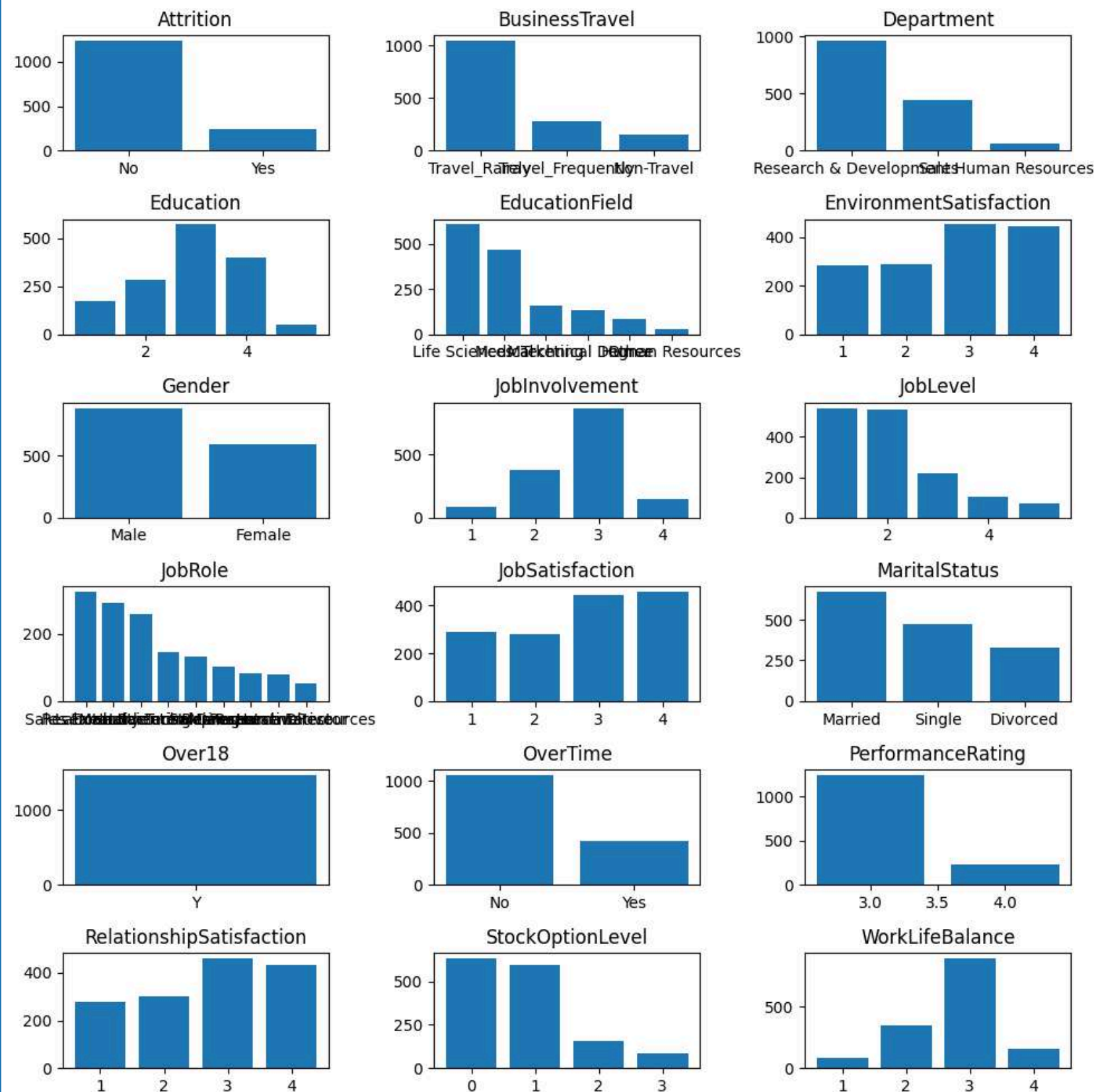
Star Schema Diagram



EDA in Python



EDA in Python

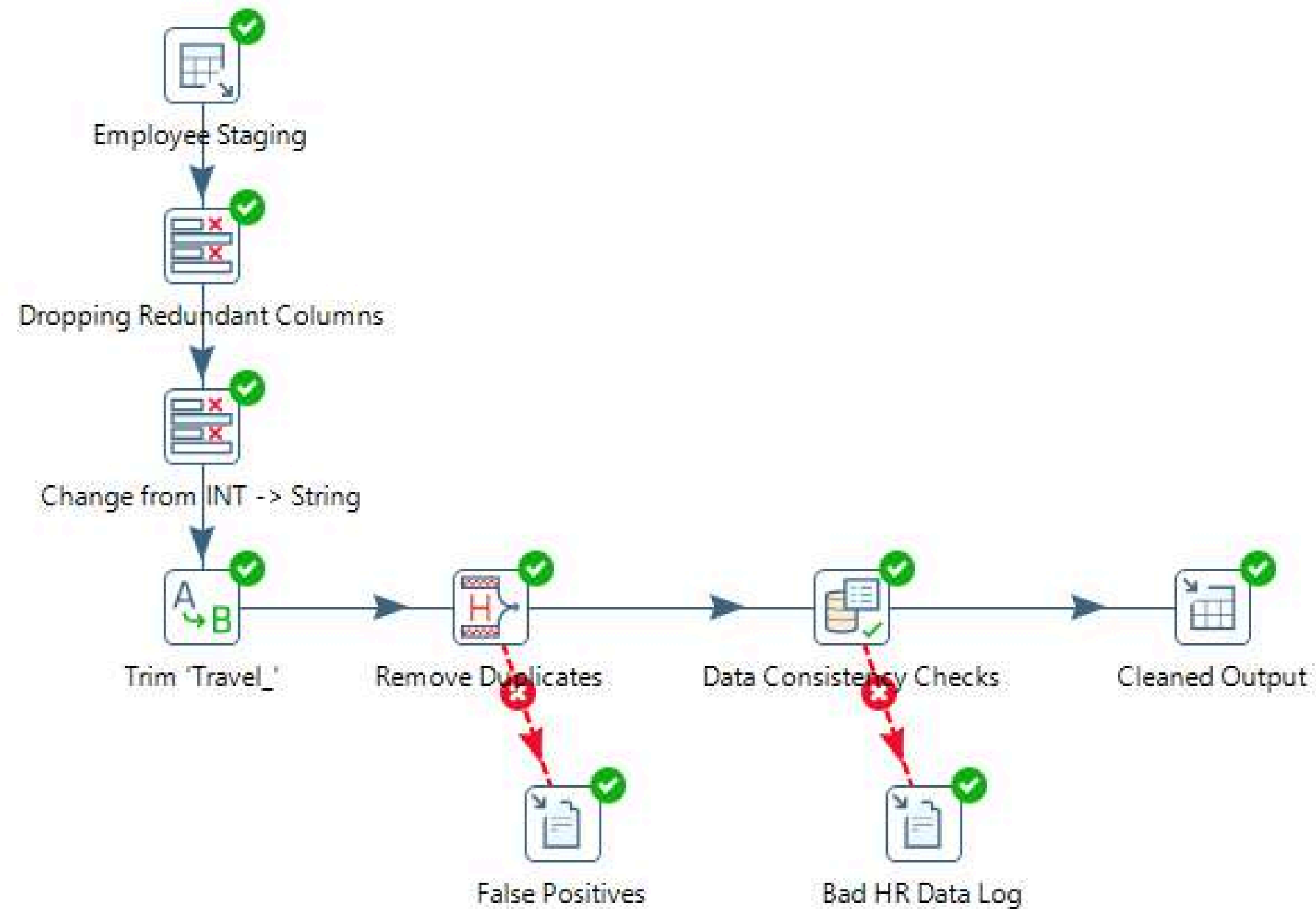


	count	mean	std	min	25%	50%	75%	max
Age	1470.0	36.923810	9.135373	18.0	30.00	36.0	43.00	60.0
DailyRate	1470.0	802.485714	403.509100	102.0	465.00	802.0	1157.00	1499.0
DistanceFromHome	1470.0	9.192517	8.106864	1.0	2.00	7.0	14.00	29.0
Education	1470.0	2.912925	1.024165	1.0	2.00	3.0	4.00	5.0
EmployeeCount	1470.0	1.000000	0.000000	1.0	1.00	1.0	1.00	1.0
EmployeeNumber	1470.0	1024.865306	602.024335	1.0	491.25	1020.5	1555.75	2068.0
EnvironmentSatisfaction	1470.0	2.721769	1.093082	1.0	2.00	3.0	4.00	4.0
HourlyRate	1470.0	65.891156	20.329428	30.0	48.00	66.0	83.75	100.0
JobInvolvement	1470.0	2.729932	0.711561	1.0	2.00	3.0	3.00	4.0
JobLevel	1470.0	2.063946	1.106940	1.0	1.00	2.0	3.00	5.0
JobSatisfaction	1470.0	2.728571	1.102846	1.0	2.00	3.0	4.00	4.0
MonthlyIncome	1470.0	6502.931293	4707.956783	1009.0	2911.00	4919.0	8379.00	19999.0
MonthlyRate	1470.0	14313.103401	7117.786044	2094.0	8047.00	14235.5	20461.50	26999.0
NumCompaniesWorked	1470.0	2.693197	2.498009	0.0	1.00	2.0	4.00	9.0
PercentSalaryHike	1470.0	15.209524	3.659938	11.0	12.00	14.0	18.00	25.0
PerformanceRating	1470.0	3.153741	0.360824	3.0	3.00	3.0	3.00	4.0
RelationshipSatisfaction	1470.0	2.712245	1.081209	1.0	2.00	3.0	4.00	4.0
StandardHours	1470.0	80.000000	0.000000	80.0	80.00	80.0	80.00	80.0
StockOptionLevel	1470.0	0.793878	0.852077	0.0	0.00	1.0	1.00	3.0
TotalWorkingYears	1470.0	11.279592	7.780782	0.0	6.00	10.0	15.00	40.0
TrainingTimesLastYear	1470.0	2.799320	1.289271	0.0	2.00	3.0	3.00	6.0
WorkLifeBalance	1470.0	2.761224	0.706476	1.0	2.00	3.0	3.00	4.0
YearsAtCompany	1470.0	7.008163	6.126525	0.0	3.00	5.0	9.00	40.0
YearsInCurrentRole	1470.0	4.229252	3.623137	0.0	2.00	3.0	7.00	18.0
YearsSinceLastPromotion	1470.0	2.187755	3.222430	0.0	0.00	1.0	3.00	15.0
YearsWithCurrManager	1470.0	4.123129	3.568136	0.0	2.00	3.0	7.00	17.0

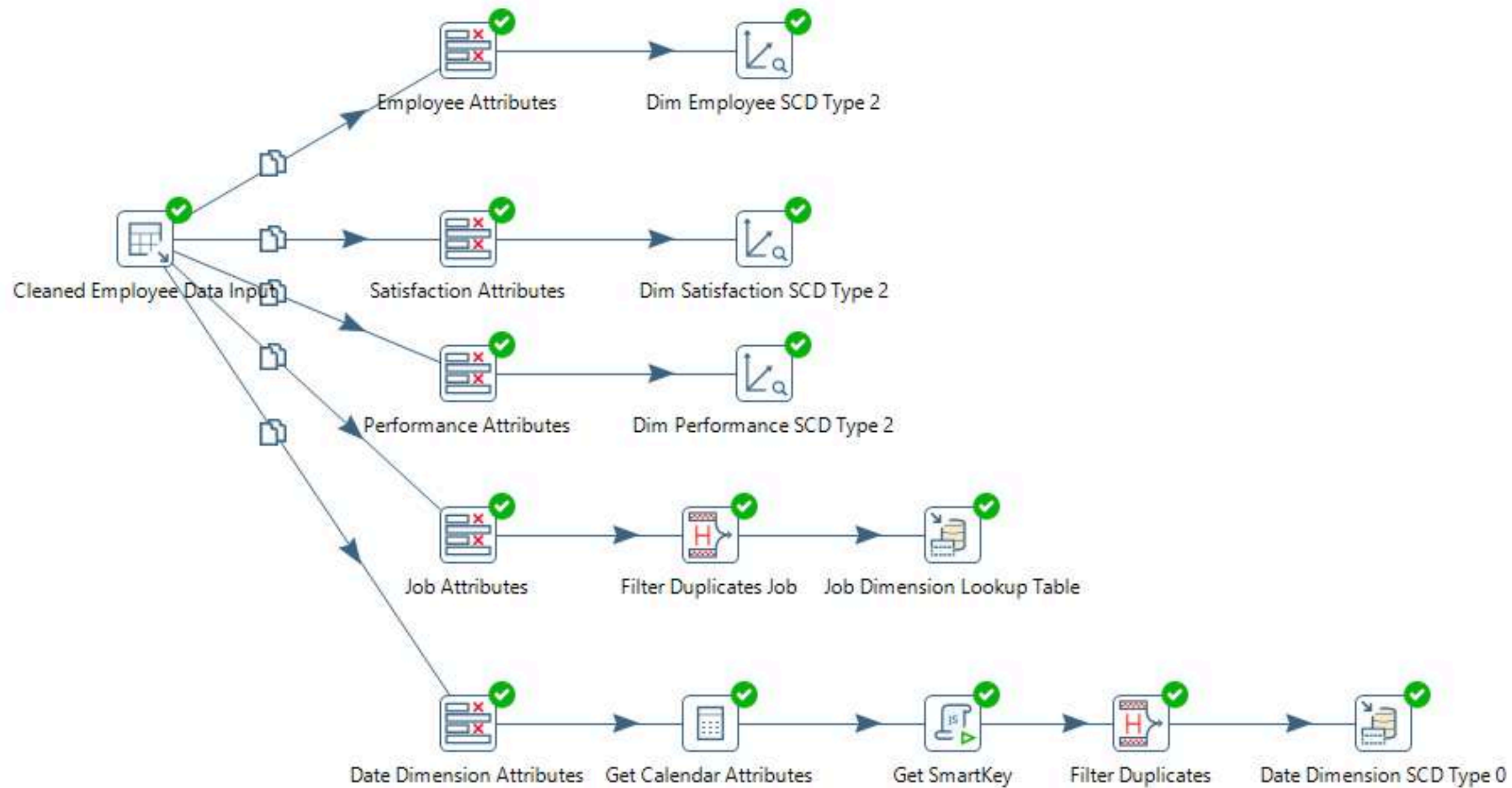
Staging Data



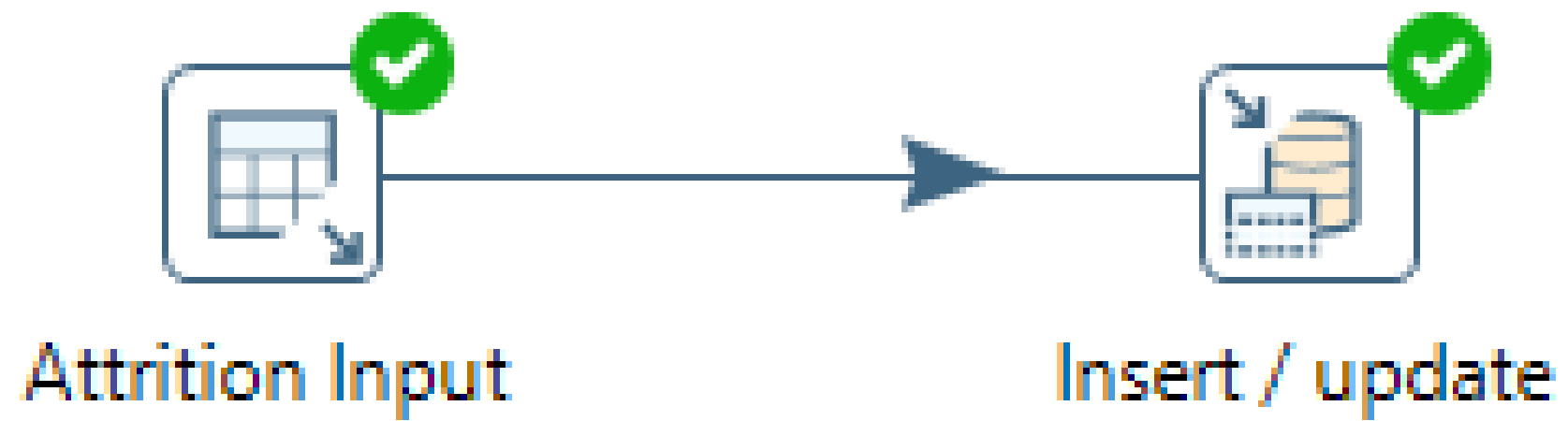
Cleaning Data



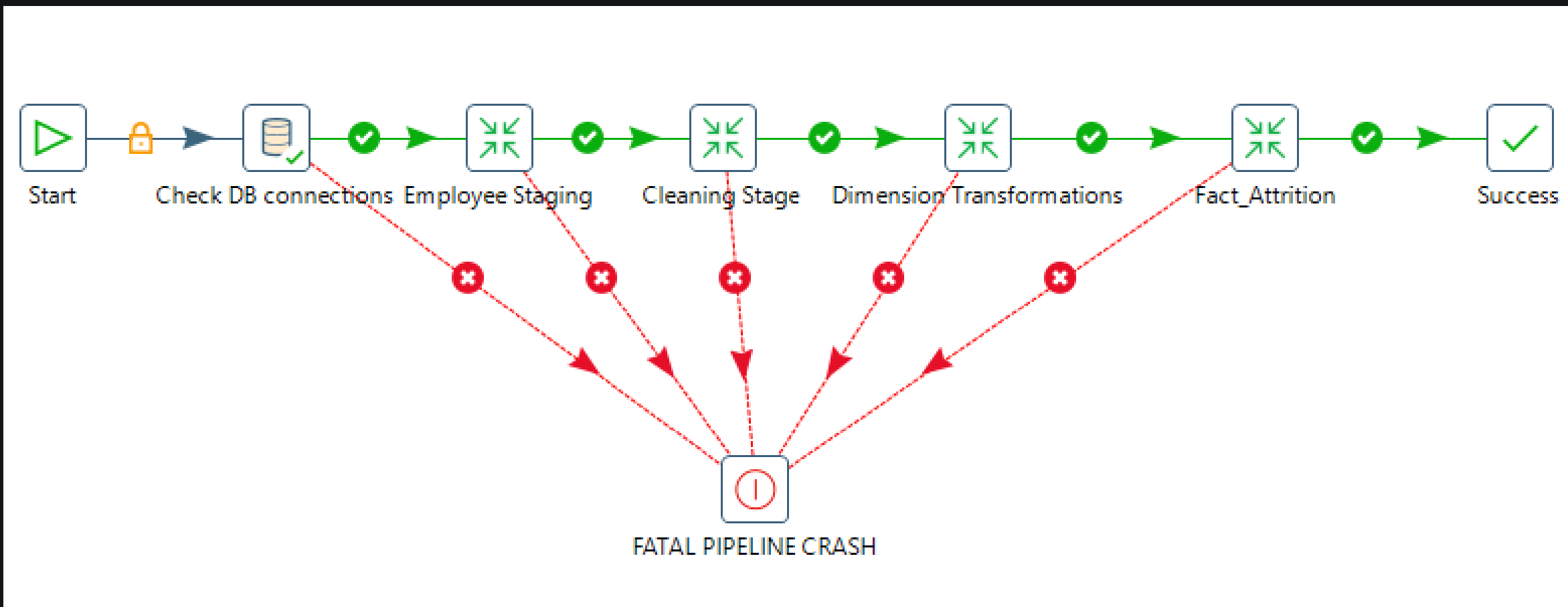
Dimension



Fact



Orchestration with Pentaho Jobs



Data Consumption (Power BI Dashboard Analytical Insight)

PowerBI Dashboard

Executive Attrition & Cost Analytics

1.47K

Total Employees

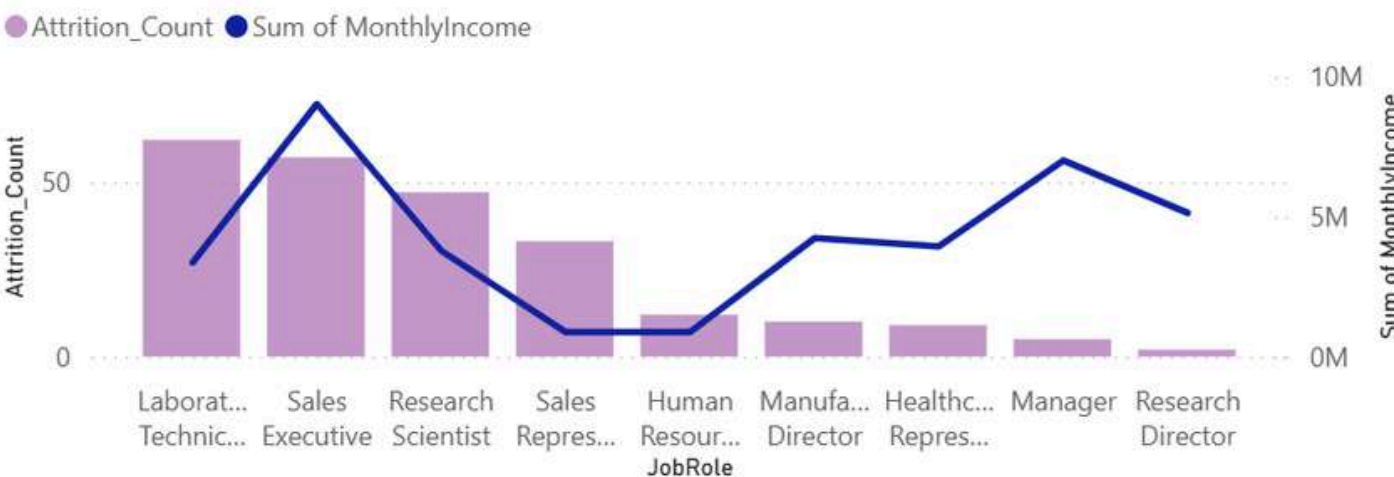
237

Attrition Yes

16.11%

Attrition_Rate

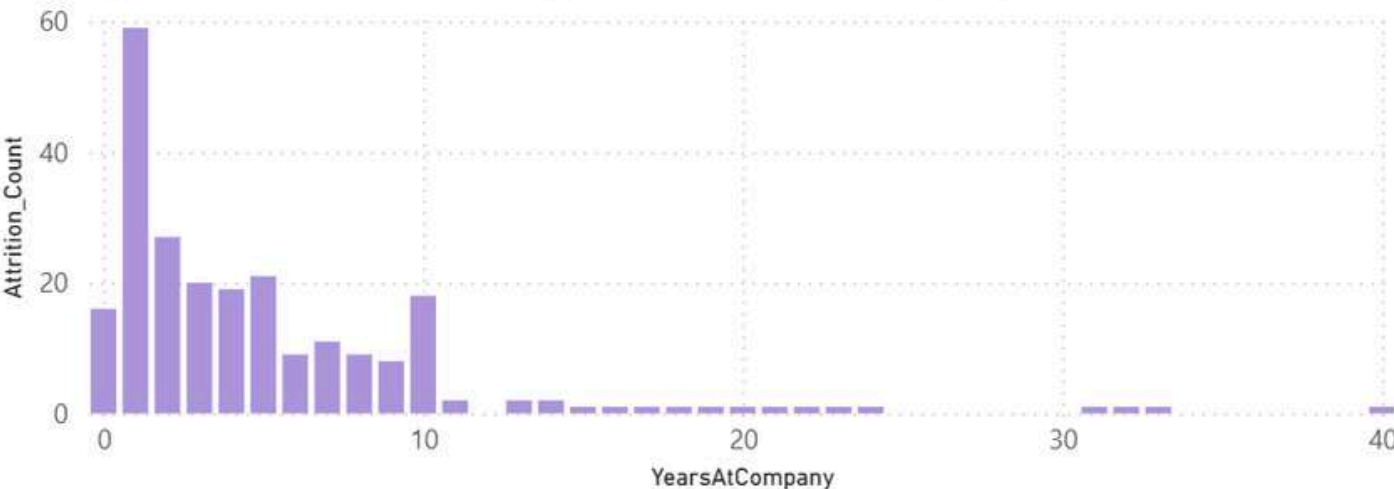
Employee Attrition Count vs. Salary Allocation by Job Role



Executive Summary: Attrition Metrics and Turnover Rates

Department	Total_Employees	Attrition_Count	Attrition_Rate
Human Resources	63.00	12	19.05%
Research & Development	961.00	133	13.84%
Healthcare Representative	131.00	9	6.87%
Laboratory Technician	259.00	62	23.94%
Manager	54.00	3	5.56%
Manufacturing Director	145.00	10	6.90%
Research Director	80.00	2	2.50%
Research Scientist	292.00	47	16.10%
Total	1471.00	237	16.11%

Employee Attrition Distribution by Tenure (Years at Company)



Payroll Budget Allocation Map across Job Roles



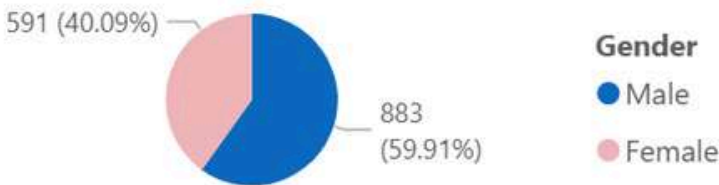
PowerBI Dashboard

Workforce Demographics & Profile

36.92

Average Workforce Age

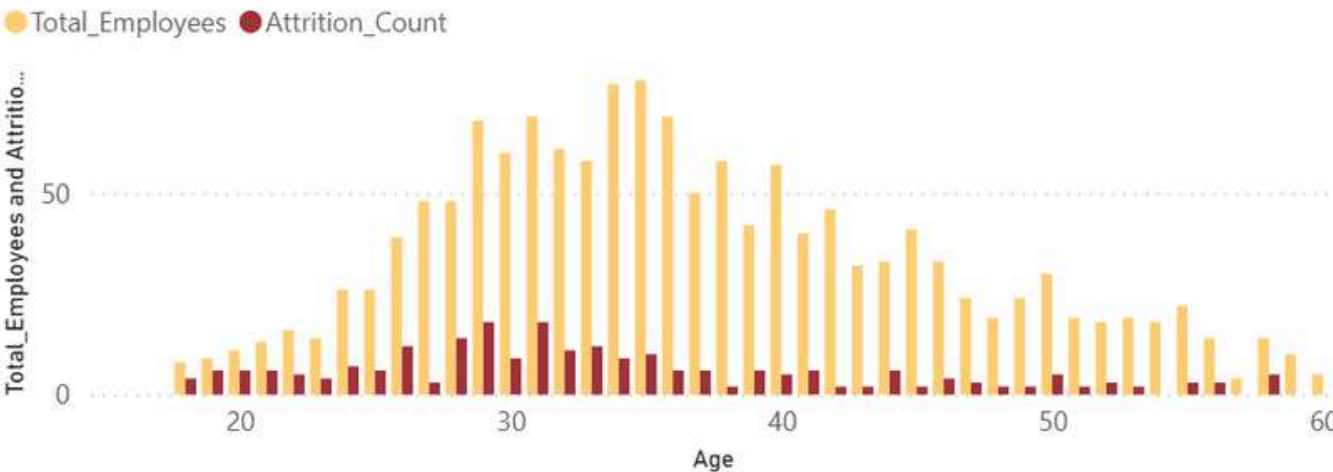
Workforce Gender Composition



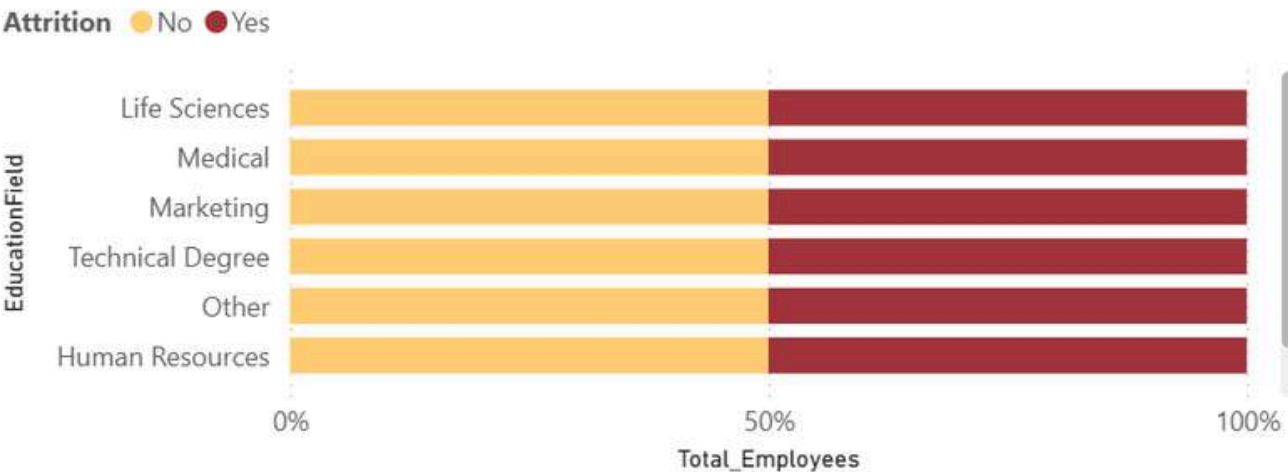
Workforce Marital Status Distribution



Active Workforce vs. Attrition Demographics by Age



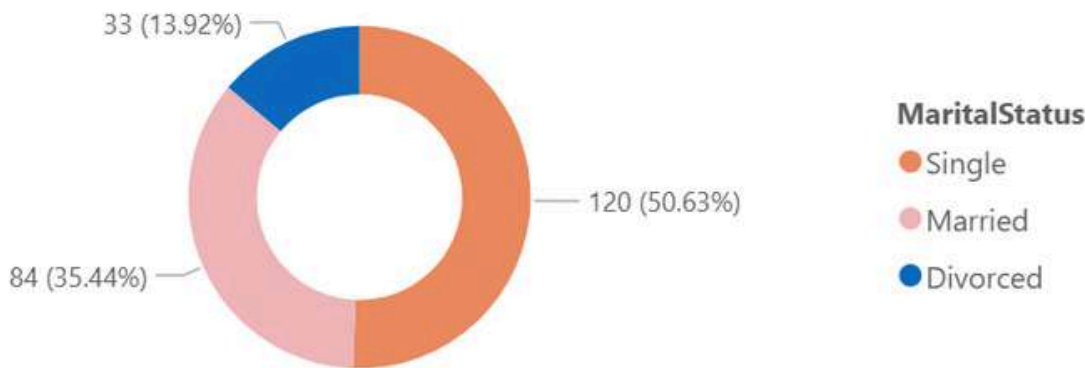
Attrition Ratio Analysis by Educational Background



Attrition Breakdown by Gender and Department

Gender	Department	Female	
		Total_Employees	Attrition_Count
+	Research & Development	379.00	43
+	Sales	189.00	38
+	Human Resources	20.00	6
	Total	588.00	87

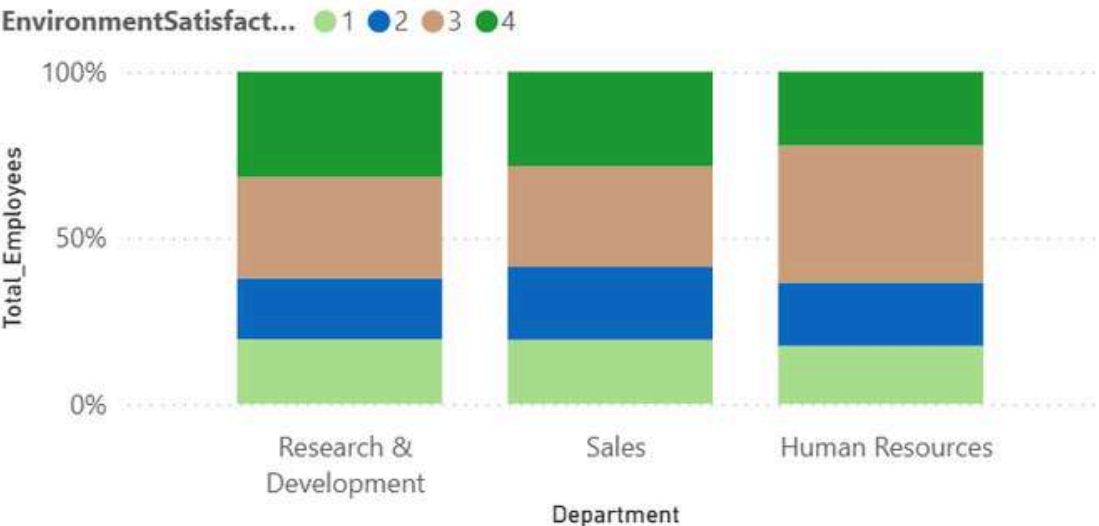
Attrition Distribution by Marital Status



PowerBI Dashboard

Employee Satisfaction & Engagement

Work Environment Satisfaction Levels per Department



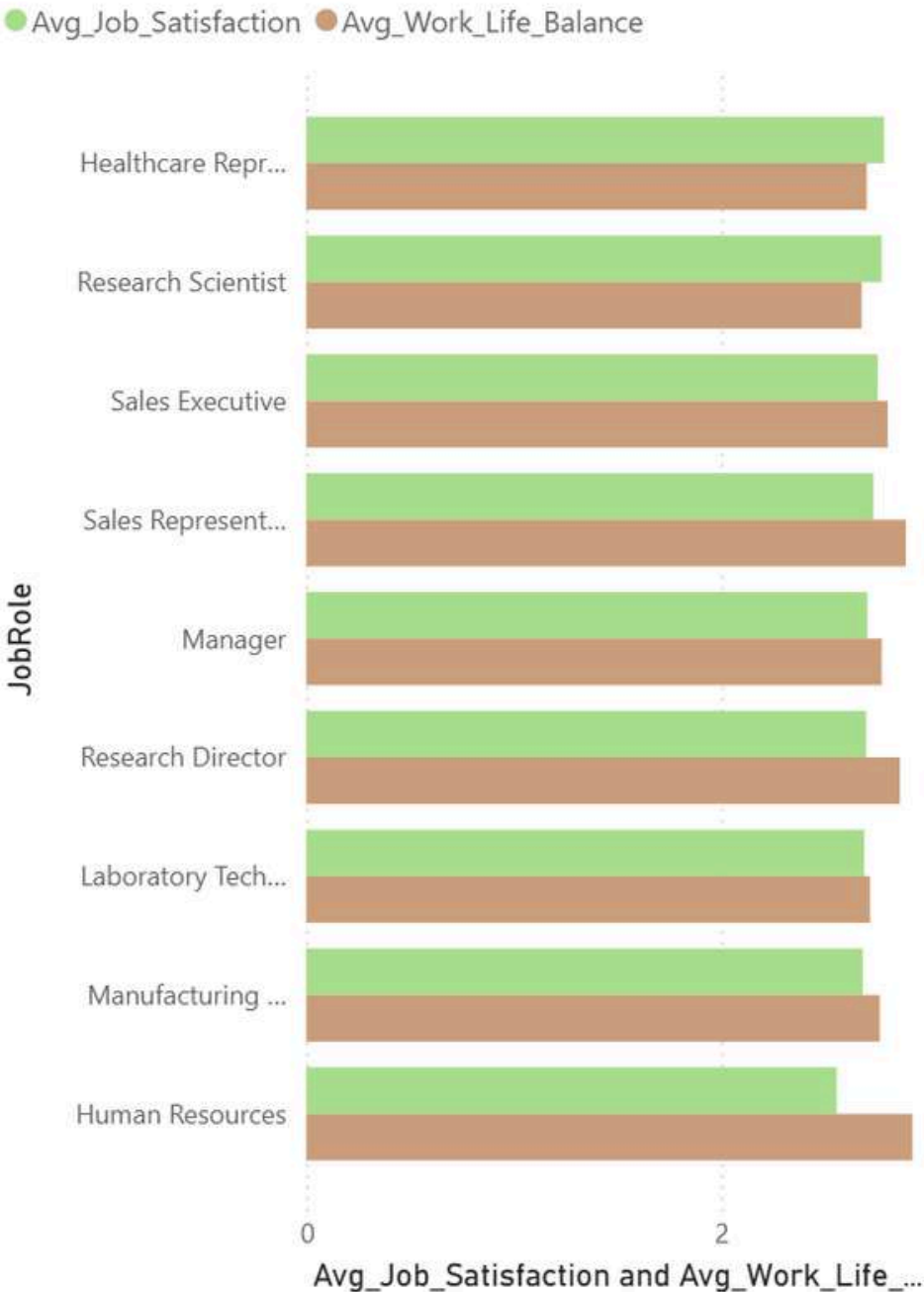
2.73

Avg Job Satisfaction

2.76

Avg Work Life Balance

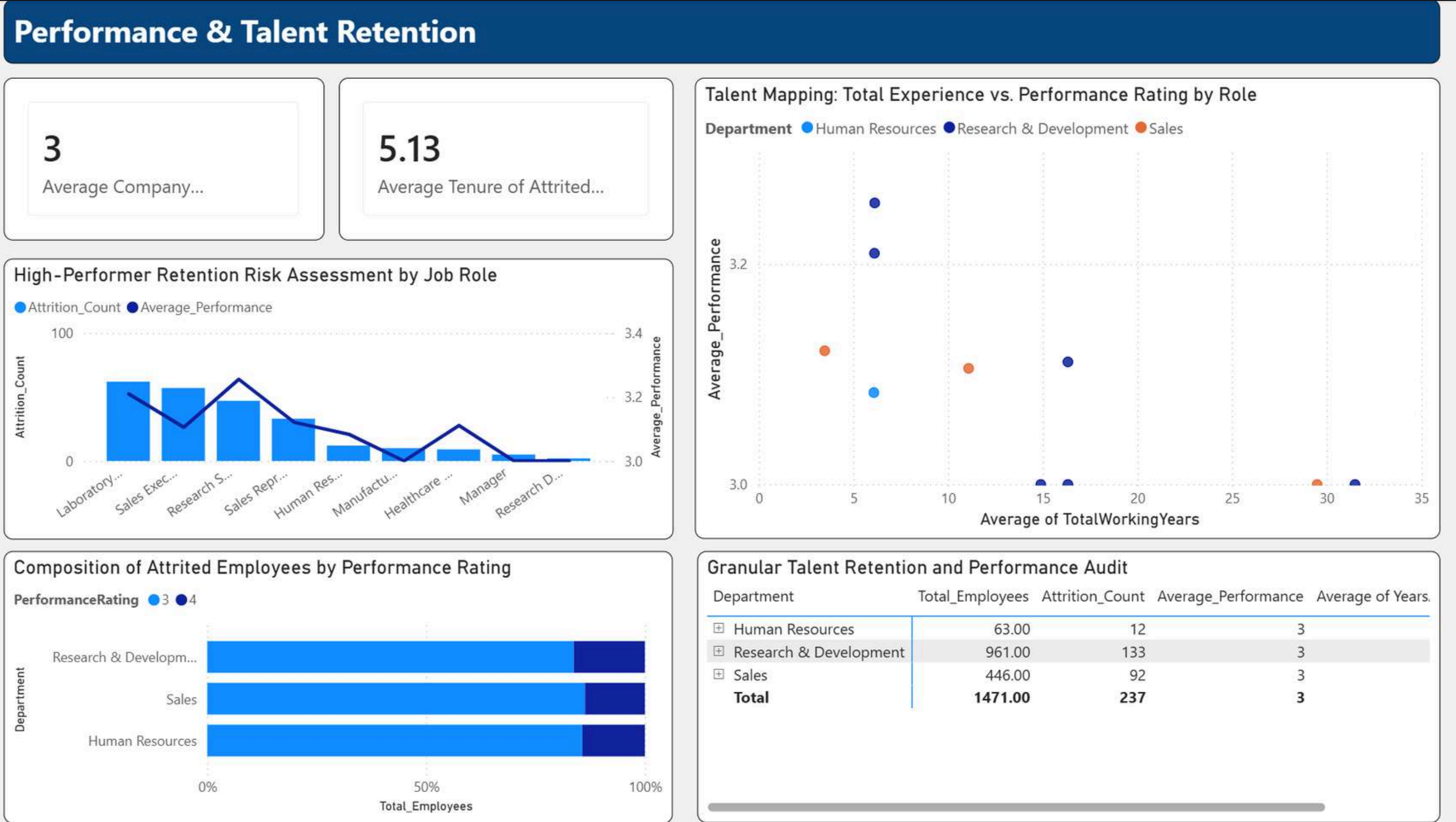
Employee Burnout Diagnosis: Job Satisfaction vs. Work-Life Balance



Employee Satisfaction Score Matrix by Job Role

Department	Avg_Job_Satisfaction	Avg_Work_Life_Balance	Avg_Environment_Satisfaction	Attrition_Rate
Research & Development	2.73	2.73	2.74	13.84%
Research Director	2.70	2.86	2.50	2.50%
Manager	2.65	2.76	2.80	5.56%
Healthcare Representative	2.79	2.70	2.77	6.87%
Manufacturing Director	2.68	2.77	2.92	6.90%
Research Scientist	2.77	2.68	2.73	16.10%
Laboratory Technician	2.69	2.72	2.72	23.94%
Human Resources	2.60	2.92	2.68	19.05%
Manager	2.82	2.91	3.09	
Human Resources	2.56	2.92	2.60	23.08%
Sales	2.75	2.82	2.68	20.63%
Manager	2.76	2.76	2.62	5.41%
Sales Executive	2.75	2.80	2.67	17.48%
Total	2.73	2.76	2.72	16.11%

PowerBI Dashboard



Thank You